Worksheet 3: Atmospheres: Structure, Clouds, & Climate

Covers Lectures 5 and 6: vertical structure, radiative balance, clouds, winds, climate

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that …"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end. All parts are analytical; no computer is needed.

Constants and data

$k_B = 1.381 \times 10^{-23}$ J K^{-1} $m_u = 1.661 \times 10^{-27}$ kg $\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4} $S_\oplus = 1361$ W m^{-2} (solar constant at 1 AU) $M_\oplus = 5.972 \times 10^{24}$ kg

Earth: $P_0 = 1.013 \times 10^5$ Pa, $R_\oplus = 6371$ km, $T_{eq} = 255$ K, surface 288 K, $\Omega = 7.29 \times 10^{-5}$ rad s^{-1}, $\Gamma_d = 9.8$ K km^{-1}

Water: $L_v = 2.50 \times 10^6$ J kg^{-1}, $R_v = 462$ J kg^{-1} K^{-1} ($L_v/R_v = 5400$ K), triple point (273 K, 611 Pa)

Body	T (K)	μ	g (m s^{-2})	Bond albedo A
Venus (0.72 AU)	737 (surface)	43.4	8.87	0.77
Earth (1 AU)	288 (surface)	28.97	9.81	0.30
Titan	94 (surface)	28.6	1.35	—

Atmospheric data as in the Lecture 5 notes; μ is dimensionless, in units of m_u. The young Sun had $L/L_\odot = 0.71$ (Lecture 6 notes).

Problem 1 How thick is an atmosphere?

The scale height sets the thickness of every atmosphere in the solar system, and hydrostatic balance connects the surface pressure to the mass sitting overhead.

(a) Compute the scale height $H = k_B T/(\mu\, m_u\, g)$ for Earth and for Titan.

(b) Titan is far colder than Earth, and its nitrogen atmosphere has nearly the same μ as air. Explain, from the structure of the scale-height formula, why Titan's atmosphere is nonetheless more than twice as extended as Earth's.

(c) Integrate hydrostatic balance $dP = -\rho g\, dz$ over the whole column to show that the surface pressure equals the weight per unit area of the overlying gas, $P_0 = m_{col}\, g$, where m_{col} is the column mass per unit area. Evaluate m_{col} for Earth, and from it the total mass of Earth's atmosphere.

Problem 2 Sunlight in, infrared out

Radiative equilibrium sets the temperature every atmosphere starts from; the greenhouse effect explains why surfaces are warmer than that.

(a) Compute the solar flux at Venus's orbit, then the equilibrium temperature of Venus,

$$T_{eq} = \left[\frac{(1-A)\,S}{4\sigma}\right]^{1/4},$$

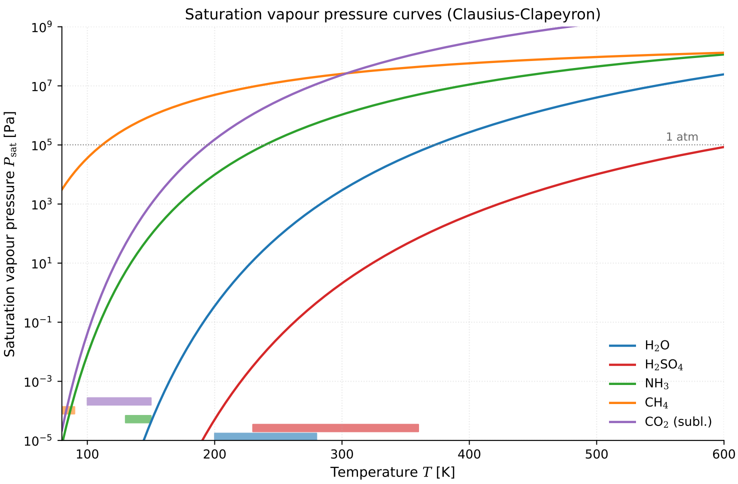
and compare it with the value in the Lecture 5 notes.

(b) Venus's measured surface temperature is 737 K, more than 500 K above the value you just computed. Explain the mechanism that maintains this difference. Your answer must name the quantity that the atmosphere blocks, and why blocking it raises the surface temperature.

(c) A fellow student writes: "back-radiation from the atmosphere is a second energy source, so the surface receives more energy than the Sun provides, and that violates energy conservation." Judge this claim and correct what is wrong with it.

Problem 3 From vapour to cloud base

The Clausius-Clapeyron equation from the blackboard derivation of Lecture 6 predicts where clouds form. The figure below reproduces the saturation curves from the notes; part (b) reads it.



(a) A parcel at the surface has $T = 293$ K and relative humidity 50%. Use the integrated Clausius-Clapeyron equation with the triple point of water as the reference state to compute its vapour pressure (show that $P_{sat}(293\text{ K}) \approx 2357$ Pa and $P_{vap} \approx 1179$ Pa), invert the Clausius-Clapeyron equation for the dew point (show that $T_d \approx 282.4$ K), and from the dry adiabatic cooling rate find the altitude of the cloud base. Take the vapour pressure as constant during ascent.

(b) Read the saturation-curve figure: a body with a solid surface has regions at 90 K where one atmospheric gas is present at a partial pressure of about 10^4 Pa (0.1 bar). Identify from the figure which of the five species is at its saturation point under these conditions, and hence which body in the solar system this is. Justify your reading in one or two sentences.

(c) The Lecture 6 notes state that Venus's sulfuric-acid cloud base is sharply defined, while Titan's methane clouds can form over a wide range of altitudes. Use the shape of the saturation curves in the figure (steep for H_2SO_4, flat for CH_4) to explain both observations with the same argument.

Problem 4 Winds on a rotating planet

Rotation organises all large-scale flow. Here you quantify it for the air above the Netherlands, then judge a claim.

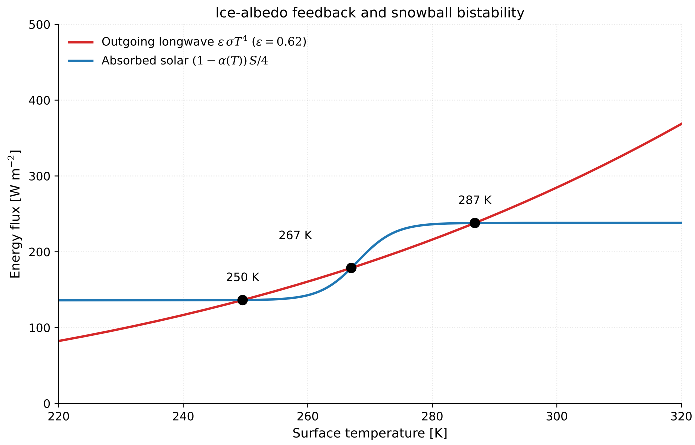
(a) Compute the Coriolis parameter $f = 2\Omega \sin\phi$ for Groningen ($\phi = 53^{\circ}$N).

(b) A low-pressure system 2000 km across has winds of 15 m s^{-1}. Compute its Rossby number and state which force balance governs the flow.

(c) A fellow student writes: "Venus rotates 243 times more slowly than Earth, so the Coriolis force is weak, and its atmosphere must therefore break up into many more small circulation cells than Earth's." Judge this claim.

Problem 5 Climate on the edge

The energy-balance figure below reproduces the ice-albedo bistability diagram from the Lecture 6 notes; you will need it in part (b).



(a) The young Sun shone at 71% of its present luminosity. Compute Earth's effective temperature at that time, taking today's albedo, and state the paradox this number creates when combined with the geological record.

(b) Read the bistability figure: identify which of the three intersections are stable and which is unstable, and describe what happens to the climate if the planet sits at the middle intersection and is nudged slightly toward warmer temperatures.

(c) A large volcanic province doubles the volcanic CO_2 outgassing rate for a few million years. Using the carbonate-silicate cycle of Lecture 6, describe the chain of responses that returns Earth's climate toward balance, and explain why this thermostat no longer works on Mars.

(d) A fellow student proposes: "the faint young Sun paradox is solved because the early Earth had more continental area than today, and rock absorbs more sunlight than water." Judge both halves of this claim against the Lecture 6 notes.